Attention as a Core Component in AI: From Naive to Efficient Implementation

ML

Supervisor/s: Chih-Jen Lin

SKILLS PREREQUISITES

- *Familiarity with matrix multiplication is enough for this project. No advanced background is required.*
- *We may write some simple C or CUDA code, but prior experience is not required. The main challenge is designing experiments and analyzing performance results.*
- *Curiosity about how modern AI systems run efficiently on hardware.*

PROJECT OVERVIEW

Attention is a core component in modern AI systems, especially large language models (LLMs). However, naive attention implementations become slow and memory-intensive as the sequence length grows. Many works aim to accelerate attention. One of the most well-known approaches is FlashAttention. While most prior methods focus on reducing computation, FlashAttention takes a different perspective: in practice, a major bottleneck is often how data is accessed and moved in memory. By reducing unnecessary memory access, FlashAttention achieves significant speed improvements. However, the original FlashAttention paper mainly presents the final algorithm and its empirical speedup, without fully explaining the reasoning behind it. It does not provide a detailed step-by-step derivation of the efficient algorithm in FlashAttention, nor does it clearly explain how memory bottlenecks are addressed. It shows great overall speedup but did not investigate if the empirical results match the algorithm design. Understanding these aspects is important. Because only by identifying the source of inefficiency and understanding how the algorithm is derived can we apply similar ideas to improve other AI models. This project will focus on the second aspect. Students will design ways to connect the FlashAttention algorithm and its empirical behavior.

SCOPE OF WORK

1. *Review FlashAttention and prior attention-acceleration methods, focusing on how FlashAttention addresses memory access and movement bottlenecks and what is (and is not) explained in the original paper.*

2. *Design ways to connect the FlashAttention algorithm and its empirical behavior.*

3. *Investigate whether empirical results match the FlashAttention algorithm design (connect algorithm behavior to empirical behavior*

EXPECTED OUTCOMES

- *Deep understanding of FlashAttention.*
- *Hands-on experience in implementing FlashAttention and conducting run-time analysis for AI models.*
- *Deep research on evaluating whether theoretical expectations match experimental results.*

FACULTY SUPERVISOR BIO

Chih-Jen Lin

Affiliated Professor of Machine Learning

Bio: *Prior to joining MBZUAI,Professor Lin is currently a distinguished professor at the Department of Computer Science, National Taiwan University. He obtained his bachelor degree from National Taiwan University, Taiwan in 1993 and Ph.D. degree from University of Michigan, USA in 1998.*

His major research areas include machine learning, data mining, and numerical optimization. He is best known for his work on support vector machines (SVM) for data classification. He and his team developed widely used machine learning packages including LIBSVM (a library for support vector machines) and LIBLINEAR (a library for large linear classification).

He has received many awards for his research work, including best paper awards in some top computer science conferences. Professor Lin is a fellow of the Association of Advancement of Artificial Intelligence (AAAI), the Institute of Electrical and Electronics Engineers (IEEE), and the Association for Computing Machinery (ACM) for his contribution to machine learning algorithms and software design.

Research Interests:

Professor Lin's main research interests are the development of machine learning algorithms and software. He bridged optimization techniques and machine learning algorithms. He pioneered the construction of open-source machine learning packages and has focused a lot on the practical realization of machine learning methodology.

Selected Publications:

[1] Li-ChungLin, Cheng-Hung Liu, Chih-Ming Chen, Kai-Chin Hsu, I-Feng Wu, Ming-Feng Tsai, and Chih-Jen On the use of unrealistic predictions in hundreds of papers evaluating graph representations. In Proceedings of the Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), 2022.

[2] Bowen Yuan, Yu-Sheng Li, Pengrui Quan, and Chih-Jen Efficient optimization methods for extreme similarity learning with nonlinear embeddings. In Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2021.

Contact:

Email: chihjen.lin@mbzuai.ac.ae

PROJECT TEAM

NAME	EMAIL
Ya-Xu Liu	Yaxu.Liu@mbzuai.ac.ae
Chin-Hui Chu	chinhuichu@gmail.com